M^2FA-Net: A Multimodal Multi-Level Fusion Attention Network for Weed Segmentation

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Abstract

Accurate weed segmentation is a fundamental requirement for precision agriculture, enabling targeted weed management and reducing excessive herbicide usage. However, reliable discrimination between crops and weeds remains challenging due to strong visual similarity, overlapping vegetation, varying illumination conditions, and complex field backgrounds. In this paper, we propose $\boldsymbol{M^2}$FA-Net, a multimodal multi-level fusion attention network for crop-weed segmentation that effectively integrates RGB imagery, multispectral information, and vegetation indices. The proposed framework employs dual DeepLabV3-ResNet50 backbones to independently extract spatial and spectral features, followed by progressive multi-level feature fusion to capture complementary information across multiple semantic scales. Convolutional Block Attention Modules (CBAM) are incorporated at each fusion stage to enhance discriminative feature learning, while deep supervision provides improved gradient propagation and optimization stability. Furthermore, a composite loss function combining Cross-Entropy, Dice, and Boundary losses is adopted to jointly optimize pixel-level classification, region overlap, and boundary accuracy. Extensive experiments conducted on the WeedyRice-RGBMS-DB dataset demonstrate the effectiveness of the proposed

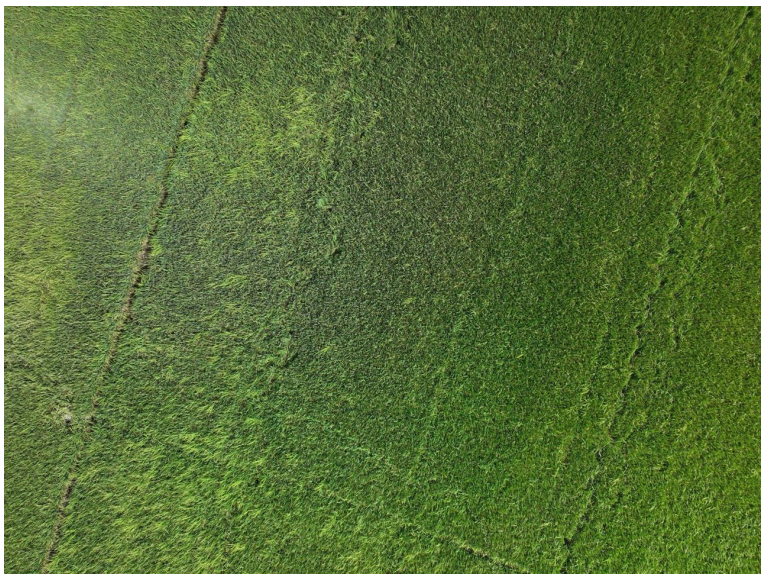


Fig. 1: Sample of RGB field image from the dataset showing the visual similarity and overlap between crops and weeds, highlighting the importance of accurate crop-weed segmentation in precision agriculture.

framework. $\boldsymbol{M^2}$FA-Net achieves a test Intersection over Union (IoU) of 80.79% and Dice score of 89.21%, outperforming eleven state-of-the-art CNN-based, transformer-based, and hybrid segmentation models. Ablation studies further validate the contribution of each architectural component. The results demonstrate that $\boldsymbol{M^2}$FA-Net provides an effective solution for crop-weed segmentation on the evaluated benchmark in UAV-based precision agriculture applications.

Keywords: Crop-weed segmentation, Precision agriculture, Multispectral imaging, Multi level fusion, CBAM, Deep supervision, Semantic segmentation, UAV imagery, Vegetation indices

1 Introduction

The global population is growing at an unprecedented rate and is estimated to reach nearly 9 billion by 2050, resulting in a significantly increased demand for food, feed, fiber, and fuel. This situation creates the need for a substantial increase in agricultural productivity [1]. However, modern agriculture faces many challenges, including decreasing arable land, climate change, and the growing problem of weeds. As a result, significant losses occur in crop production, and weeds can reduce the productivity of crops by 25%. This means effective weed management is an important component of sustainable agriculture. This problem becomes more serious in rice fields as weedy rice is morphologically very similar to conventionally cultivated rice

during the early growth stages, making it difficult to distinguish. As illustrated in Fig. 1, even visual inspection of field images often makes it challenging to clearly distinguish crops from weeds due to overlapping vegetation, similar texture, and comparable color appearance. This morphological similarity makes visual identification difficult and labor-intensive, which often results in delays in intervention or reduced effectiveness[2][3]. Conventional weed management methods, such as manual inspection and uniform herbicide spraying, are not only inefficient but also unsustainable from an environmental perspective due to increased chemical usage.

To tackle these issues, precision agriculture is increasingly using unmanned aerial vehicles (UAVs) alongside deep learning methods for the automated differentiation of crops and weeds. Imaging via UAVs allows for the capture of high-resolution images of agricultural fields, offering detailed spatial data crucial for identifying weeds at an early stage [4–9]. In this context, models based on Convolutional Neural Networks (CNNs), including *YOLO*[10], *Mask R-CNN* [11], *U-Net*[12], *SegNet* [13], and *DeepLab*[14][15], have shown excellent results in tasks involving semantic segmentation. For instance, Silva et al. [16] presented a comparison among different variants of YOLO, U-Net, and Mask R-CNN, where YOLOv8s achieved more than 99.7% precision on publicly available aerial images of soybeans and beans.

CNNs have demonstrated good performance in segmentation tasks by exploiting local spatial features, but they still face challenges in complex scenarios, such as agricultural crop-weed segmentation. This is because they cannot effectively model the long-range contextual dependencies that are crucial in complex agricultural scenes. To address this challenge, attention-based approaches, such as the Convolutional Block Attention Module (CBAM), have emerged as effective solutions [17–20]. These mechanisms allow the models to learn long-range dependencies and improve the extraction of global contextual information. For instance, Li et al.[21] have shown 3.08% improvement on UNet architecture by leveraging attention mechanisms on the sugar beet dataset. Besides, Transformer-based methods [22–24] are proposed to further improve the modeling of global context. However, the application of these methods in real-time UAV-based applications is still hampered by two main limitations, namely, the requirement of large-scale training datasets and the high computational complexity. For instance, Jiang et al.[25] explore different variants of transformer on their in-house weed datasets and report that SegFormer achieves the best mIoU 65.74% compared to all other variants, ie Swin Transformer. Another important trend in crop–weed segmentation is the fusion of vegetation indices [26] and multispectral imaging [27] with RGB data. Multispectral images provide additional spectral information about plant physiology, while vegetation indices characterize features such as chlorophyll content and vegetation vigor that are important for enhancing the separability of crops from weeds. Ma et al. [28] have shown that the integration of these modalities into deep learning frameworks could improve the segmentation performance. However, most existing methods use simple early fusion strategies, which cannot effectively solve the distribution shift problem from heterogeneous data modalities.

Existing approaches offer distinct advantages and also have certain limitations in segmenting weedy rice under challenging field conditions. CNN encoder-decoders such as U-Net and DeepLabV3 capture local texture and multi-scale context through atrous

convolution, but their receptive fields are limited, and reasoning across overlapping plants is difficult. Attention modules such as CBAM can adjust the channel and spatial locations at a low cost, but can not enhance the features extracted by the backbone. Transformer-based models can capture global context but require large annotated datasets and heavy computation, which is unaffordable for small UAV datasets and field deployment. Multispectral bands and vegetation indices can add physiological cues that RGB lacks [28], but they are usually combined with RGB by stacking channels at the input [29, 30]. This may force one set of early filters to model modalities with different statistics. There is no single approach that addresses all these limitations.

Hence, we propose **M^{2}FA-Net**, a Multi-Level Multimodal Fusion framework for crop-weed segmentation. The proposed method performs hierarchical feature fusion at multiple progressive encoder levels, enabling the model to capture complementary cross-modal cues at multiple spatial resolutions and abstraction levels. CBAM attention modules are applied after each fusion level to selectively recalibrate the integrated features. Deep supervision via auxiliary classifiers provides rich intermediate gradient signals, and a composite loss function jointly optimizes pixel accuracy, region overlap, and boundary sharpness. Specifically, features from two modality-specific encoders are fused at each of the four encoder stages using concatenation followed by a learned 1×1 convolution, while each input modality is evaluated separately.

Although individual blocks are well established, the novelty of M^2FA-Net lies in its integration, which differentiates it from existing RGB-multispectral approaches in 3 key aspects.

1. *Fusion location:* Two modality-specific encoders are fused at every encoder stage rather than at the input or at a single late stage.
2. *Information interaction:* At each stage, concatenated features are combined by a learnable 1×1 convolution and then refined through CBAM, enabling interaction between the modalities at every scale.
3. *Feature selection:* CBAM utilizes channel attention to assign different weights to RGB, multispectral, and vegetation indices, while spatial attention identifies the most informative regions. Auxiliary classifiers are applied to intermediate fused features that provide direct gradient feedback to each fusion stage.

The main contributions of this work are summarized as follows:

- Proposed a dual-stream DeeplabV3 architecture pretrained on ResNet50 that fuses RGB and multispectral features at multiple hierarchical encoder levels, enabling adaptive integration of complementary cross-modal information across all spatial resolutions.
- Integrated CBAM attention modules at each fusion stage to selectively emphasize discriminative channel and spatial features, suppressing background noise and improving cross-modal feature quality at every level of abstraction.
- Introduced a deep supervision mechanism via multiple auxiliary segmentation classifiers attached to intermediate fusion stages, providing dense gradient flow to all backbone layers and substantially improving convergence and final segmentation performance.

- Adopted a composite loss function that combines cross-entropy, Dice, and boundary-sensitive loss terms, jointly optimizing pixel-level accuracy, region overlap, and boundary sharpness for challenging crop-weed segmentation.
- Experimented on the WeedyRice-RGBMS-DB dataset and achieved best performance among the evaluated methods, with a test IoU of 80.79% and a Dice score of 89.21%, surpassing eleven competitive baseline models.
- Conducted an input-modality study to quantify the contributions of multispectral bands and individual vegetation indices to the model performance.

2 Related Works

Agricultural image analysis has made remarkable progress with the rise of deep learning, enhancing crop monitoring, disease diagnosis, weed detection, and semantic segmentation tasks significantly [31]. We are moving away from handcrafted feature-based methods that have been widely used in previous studies, as such approaches often fail in real-world conditions with illumination variation, occlusion, and complex backgrounds. Weed detection and segmentation remain particularly challenging tasks due to the visual similarities between crops and weeds, varying lighting conditions, and different soil backgrounds [4, 5]. Over time, researchers have explored a wide spectrum of strategies, ranging from convolutional networks to attention-enhanced models and transformer-based frameworks, as well as multimodal data fusion using both RGB and multispectral images, enabling the extraction of vegetation indices such as NDVI and NDRE.

2.1 CNN-Based Segmentation Approaches

Convolutional Neural Networks laid the foundation of semantic segmentation in precision agriculture. Encoder-decoder architectures such as U-Net [12], U-Net++ [32], PSPNet [33], HRNet [34], DenseUNet [35], and DeepLabV3+ [15] have been widely adopted for UAV-based crop monitoring and weed mapping, demonstrating strong ability to capture local spatial patterns and contextual scene information [16]. Among these, DeepLabV3+ has drawn attention for its use of atrous convolutions and Atrous Spatial Pyramid Pooling (ASPP), which increase the receptive field with minimal extra computation, making it suitable for dense prediction tasks like weed segmentation[15]. To further improve performance, attention mechanisms such as the Convolutional Block Attention Module (CBAM)[17] have been introduced into CNN pipelines, jointly modeling channel and spatial attention to improve localization accuracy and boundary precision in agricultural scenes[29]. Beyond architectural choices, boundary-aware loss functions and deep supervision strategies have also proven effective in producing improved segmentation outputs, especially where precise plant contour delineation matters for targeted herbicide application.

Recent studies have also explored lightweight deep learning models for weed identification on resource-constrained devices. Li et al. [36] proposed WAGENet, a hardware-aware lightweight network for weed classification on low-power microcontrollers. Zhang et al. [37] introduced a TinyML-enabled IoT framework that employs

knowledge distillation to improve the efficiency of on-device weed classification. Similarly, Zhang et al. [38] developed TinyWeedNet, a hardware-efficient CNN for real-time weed identification on embedded devices. These studies demonstrate the importance of computational efficiency for practical agricultural applications. However, they primarily address image-level weed classification, whereas the present work focuses on pixel-level segmentation using multimodal RGB and multispectral imagery.

2.2 Transformer-Based Segmentation Approaches

More recently, transformer-based models have been introduced in the field of agricultural segmentation, which provide a completely different paradigm through global self-attention mechanisms. Architectures such as Vision Transformer (ViT) [22], SegFormer [23], Segmenter [39], and Swin Transformer [24] build on the transformer architecture introduced by Vaswani et al. [40]. These have achieved powerful segmentation performance by capturing long-range spatial dependencies that CNNs find intrinsically difficult to capture. This global context awareness is especially helpful in cluttered agricultural scenes where local features alone may not be sufficient to distinguish visually similar plant species. Besides supervised learning paradigms, alternative learning methods for agricultural vision tasks have also been explored in recent research. Kumar et al. [41] proposed a semantic attribute-based zero-shot learning framework for plant disease classification and proved that attribute-guided representations could improve the recognition of unseen disease categories. Such approaches are indicative of the growing curiosity in developing more data-efficient learning methods for agricultural applications. However, a major limitation of pure transformer models is their heavy dependence on large annotated datasets and their high computational requirements, which are practical barriers to real-time deployment in field-based agricultural systems.

2.3 Hybrid and Multimodal Fusion Approaches

Recognizing that CNNs and transformers have complementary strengths, hybrid frameworks have been proposed to combine both of these architectures so that local feature extraction and global contextual modeling can be done together and yield improved segmentation robustness [32]. Alongside architectural hybridization, multimodal data fusion has emerged as another direction. RGB imagery alone often falls short under challenging conditions such as changing illumination and background conditions, prompting researchers to integrate multispectral bands - particularly Near Infrared (NIR) and Red Edge (RE) — highly correlated with plant physiology [27]. Sahin et al. [42] fused RGB and multispectral inputs within a CRF-enhanced U-Net framework and reported meaningful segmentation gains on sunflower weed datasets, while Ma et al. [28] showed that adding vegetation indices to multispectral data improves robustness under different field conditions. Although early concatenation remains a common fusion strategy [29, 30], recent studies have explored attention-based and multi-scale cross-modal fusion to better integrate heterogeneous spatial and spectral information. Recent studies have investigated advanced multimodal fusion strategies for agricultural segmentation and remote sensing. Fan et al. [43] proposed

Table 1: Comparison of recent multimodal fusion approaches for agricultural segmentation and remote sensing.

Method	**Input Modalities**	**Fusion Strategy**	**Architectural Characteristics**
CMFNet (2024)[43]	RGB + NIR	Cross-modal multi-scale feature fusion	Cross-modal enhancement and attention-based feature refinement for weed segmentation
Cross Mamba Fusion Network (2025)[46]	Hyperspectral + LiDAR	Global–local feature extraction with Cross-Mamba fusion	Mamba-based multimodal interaction for remote-sensing classification
ASVLB-Net (2026)[44]	Multispectral + NDVI	NDVI-guided adaptive feature fusion	Lightweight architecture for spectral and vegetation feature integration
SCG-UNet (2026)[45]	RGB + Multispectral	Channel–spatial attention and attention-guided skip fusion	U-Net-based architecture for weed segmentation
M^2FA-Net (Proposed)	**RGB + Multispectral + NDVI/NDRE**	**Four-stage hierarchical fusion with 1×1 convolution and CBAM**	**Independent modality encoders, progressive fusion, and deep supervision**

CMFNet, which integrates RGB and near-infrared imagery through cross-modal feature interaction and attention-based refinement for field weed mapping. Dong et al. [44] introduced ASVLB-Net, employing NDVI-guided adaptive fusion for multispectral weed segmentation. Similarly, SCG-UNet[45] combines UAV RGB and multispectral imagery using channel-spatial feature enhancement and attention-guided skip connections for soybean weed segmentation. In remote sensing, Cross Mamba Fusion Network[46] employs Cross-Mamba modules to integrate hyperspectral and LiDAR features for classification.

Although these methods demonstrate the effectiveness of multimodal feature integration, they differ in their fusion architectures, input modalities, and learning objectives. The proposed M^2FA-Net employs independent RGB and multispectral encoders with hierarchical feature fusion at four encoder stages. At each stage, concatenated features are integrated through a learnable 1×1 convolution and refined using CBAM, while deep supervision supports intermediate feature learning. Table 1 summarizes the architectural differences between these approaches and the proposed framework.

Although recent studies have reported improved crop-weed segmentation performance using modified encoder-decoder and ensemble architectures [47], most approaches remain dependent on RGB imagery and do not fully exploit complementary multispectral information. Despite progress in these areas, several challenges remain

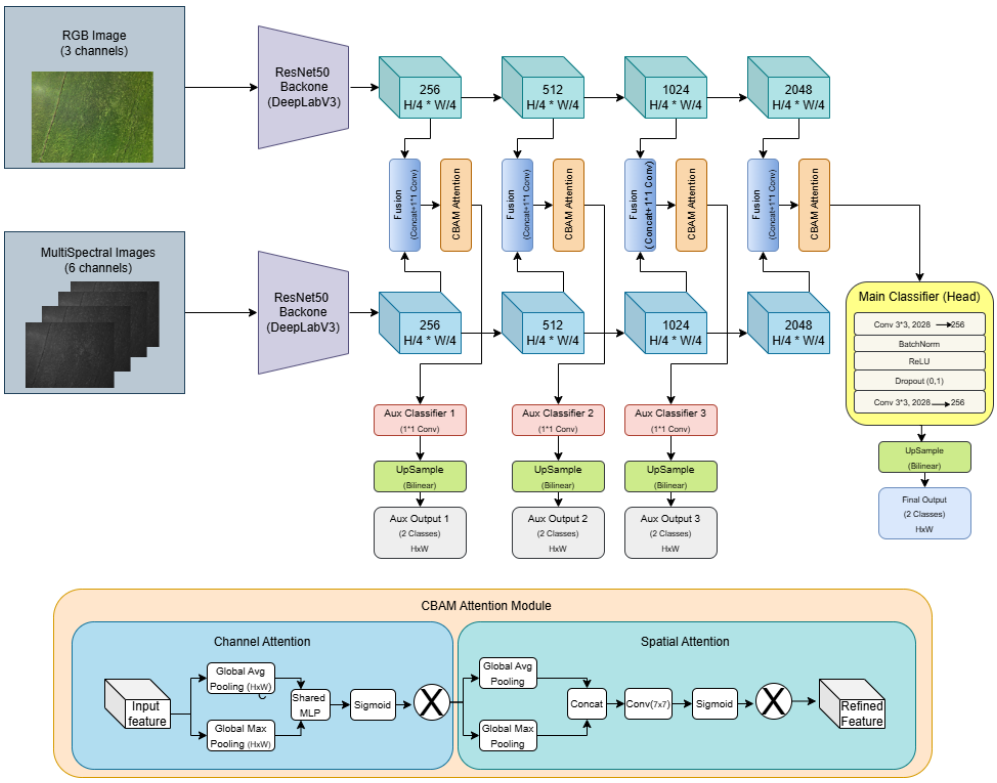


Fig. 2: Architecture of the proposed M^2FA-Net for multimodal weed segmentation.

unresolved. Most CNN-based algorithms rely only on RGB imagery and do not benefit from the full range of spectral information available with multispectral sensors. Transformer-based approaches, while powerful in modeling global context, demand substantial annotated data and computational resources that are often impractical in real agricultural deployment scenarios. Hybrid fusion methods have made strides in combining spatial and spectral modalities, yet they predominantly employ simplistic early fusion strategies that fail to capture the complementary nature of heterogeneous data streams at multiple representational levels. Furthermore, relatively few studies have brought together vegetation indices, hierarchical multimodal feature fusion, lightweight attention mechanisms, deep supervision, and boundary-aware optimization within a single cohesive segmentation framework. These collective gaps motivate the present work, which proposes a multi-level fusion DeepLab-based architecture integrating dual-stream ResNet50 backbones, CBAM attention, and a composite loss function designed to deliver accurate and robust weed segmentation under complex, real-world agricultural conditions.

3 Methodology

In this work, we propose M^2FA-Net, a multimodal attention-driven semantic segmentation framework for accurate weed segmentation using RGB and multispectral imagery. Figure 2 shows the proposed architecture and it combines spatial information from RGB images with spectral and vegetation-specific information extracted from multispectral bands and vegetation indices. The overall architecture consists of two parallel DeepLabV3-ResNet50 backbone networks: one processes RGB images while the other processes multispectral inputs containing spectral bands and vegetation indices. Unlike conventional late-fusion approaches that combine modality-specific representations only at the final encoder stage, the proposed method fuses feature maps extracted at different semantic levels progressively using concatenation and 1×1 convolution operations, enabling the integration of complementary spectral-spatial cues at every scale of abstraction. At each fusion level, a CBAM attention module recalibrates the fused representation[18, 20]. The final fused representation is passed through a segmentation head to generate dense pixel-wise weed predictions. Additionally, deep supervision and a composite loss function are incorporated to improve optimization stability, gradient flow, and segmentation boundary quality [48].

3.1 Input Modalities

Our framework considers two primary complementary input modalities:

- **RGB Images:** Standard three-channel RGB images that capture spatial structures, texture patterns, and color information.
- **Multispectral Data:** Four spectral bands, namely Green (G), Red (R), Red Edge (RE), and Near-Infrared (NIR), which provide important spectral and physiological information related to vegetation characteristics.

To further improve crop-weed discrimination, two vegetation indices are computed from the multispectral bands.

3.1.1 Normalized Difference Vegetation Index (NDVI)

NDVI is expressed in the equation 1.

$$\text{NDVI} = \frac{NIR - R}{NIR + R} \tag{1}$$

NDVI [49] highlights vegetation health by exploiting the strong reflectance of healthy vegetation in the NIR spectrum and absorption in the red spectrum.

3.1.2 Normalized Difference Red Edge Index (NDRE)

NDRE is defined in the equation 2.

$$\text{NDRE} = \frac{NIR - RE}{NIR + RE} \tag{2}$$

NDRE [50] is more sensitive to chlorophyll concentration and plant stress, making it highly effective for precision agriculture applications. Consequently, the multispectral representation is extended into a six-channel input consisting of {G, R, RE, NIR, NDVI, NDRE}. Per-channel mean and standard deviation statistics are computed over the training set and used to normalize the multispectral input using Z-scores. This multimodal representation provides complementary spatial, spectral, and vegetation-specific information for robust weed segmentation.

3.2 Network Architecture

M^2FA-Net adopts a dual-stream multi-level fusion strategy based on DeepLabV3 with ResNet50 backbones. The architecture is specifically designed to preserve modality-specific characteristics while reliably integrating spatial and spectral information.

3.2.1 Dual-Stream Multi-Level Feature Extraction and Fusion

We employ two independent DeepLabV3-ResNet50 encoders pretrained on the COCO-with-VOC-labels benchmark: an RGB encoder for three-channel RGB image, and a multispectral encoder for six-channel multispectral input comprising spectral bands and vegetation indices. As DeepLabV3 is originally designed for three channel input, the RGB encoder retains its original architecture, processes the three channel input without modification, while the first convolutional layer of the multispectral encoder is modified to accept six input channels. For the main experiments, the first three multispectral input channels inherit the corresponding pretrained RGB convolutional weights, while the remaining three channels are initialized using their channel-wise mean. In the separate input-modality experiments, all multispectral input channels are initialized using the mean pretrained RGB weights.

At each of the four encoder stages $i = 1, \ldots, 4$, the encoder outputs feature maps with progressively increasing channel dimensions and decreasing spatial resolutions, capturing the information at different semantic levels: low-level features represents edges, textures, and local structures; mid-level features represents plant structures and shape information; and high-level features capture semantic representations relevant to crops and weed identification. The extracted feature maps from the RGB and multispectral branches are stored independently for subsequent hierarchical fusion. Unlike the traditional late fusion approaches that integrates modalities at the final layers, the proposed approach performs fusion at every encoder stage to extract complementary information across multiple feature levels. This helps the model to utilize cross-modal interactions through out the feature levels. At each encoder stage, the RGB and multispectral feature maps are combined along the channel dimension and is expressed in the equation 3.

$$F_{fusion}^{(i)} = \text{Concat}(F_{rgb}^{(i)}, F_{ms}^{(i)}) \tag{3}$$

where $F_{rgb}^{(i)}$ and $F_{ms}^{(i)}$ are RGB and multispectral features at the i^{th} encoder level, respectively. After concatenation, a 1×1 convolution is applied to reduce dimensionality and learn optimal feature interactions:

$$F_{reduced}^{(i)} = \text{Conv}_{1\times1}(F_{fusion}^{(i)}) \tag{4}$$

This step integrates diverse spatial and spectral features without increasing the network's channel depth [51].

3.2.2 CBAM Attention Module

To enhance feature representation further, the Convolutional Block Attention Module (CBAM) [17] is applied to the fused features at each level. CBAM sequentially applies channel attention and spatial attention.

Channel Attention:

This attention identifies the most informative feature channels by learning inter-channel relationships and is expressed by Eq. 5.

$$F_c^{(i)} = M_c(F_{reduced}^{(i)}) \otimes F_{reduced}^{(i)} \tag{5}$$

where M_c denotes channel attention and $\otimes$ represents element-wise multiplication.

Spatial Attention:

This attention focuses on important spatial regions within the feature map and is defined in Eq. 6.

$$F_s^{(i)} = M_s(F_c^{(i)}) \otimes F_c^{(i)} \tag{6}$$

where M_s represents spatial attention.

The final attention-refined representation is given in Eq. 7.

$$F_{att}^{(i)} = F_s^{(i)} = \text{CBAM}(F_{reduced}^{(i)}) \tag{7}$$

The attention mechanism enables the network to suppress irrelevant background information, such as soil and shadows, while emphasizing discriminative weed regions.

3.2.3 Deep Supervision

To facilitate gradient flow and stabilize training, auxiliary classifiers are attached to intermediate feature levels. These auxiliary outputs provide additional supervision during training and encourage meaningful feature learning at different semantic scales. The L_{final} is expressed in the equation 8

$$L_{\text{final}} = L_{\text{main}} + \lambda \sum_{l=1}^{N} L_{\text{aux}}^{l} \tag{8}$$

where L_{main} represents the primary segmentation loss computed from the final output prediction, while L_{aux}^{l} denotes the auxiliary losses obtained from intermediate supervision levels. The weighting factor λ controls the contribution of auxiliary losses, and N represents the total number of supervised intermediate levels in the multi-level architecture.

3.2.4 Segmentation Head

The last fused feature representation is passed through a segmentation head consisting of convolution, batch normalization, ReLU activation, and dropout layers. The final prediction layer generates pixel-wise logits corresponding to the weed and background classes. The low-resolution segmentation map is upsampled to the original image resolution using bilinear interpolation and expressed in equations 9 10.

$$F_{\text{seg}} = \text{Conv}_{1\times1}\left(\text{Conv}_{3\times3}\left(F_{att}^{(4)}\right)\right) \tag{9}$$

$$F_{\text{out}} = Upsample(F_{\text{seg}}) \tag{10}$$

where F_{seg} represents the predicted segmentation map before upsampling.

3.3 Loss Function

The model is trained using a composite loss function combining three complementary terms to improve segmentation quality and boundary precision.

3.3.1 Cross-Entropy Loss

Cross-Entropy Loss supervises pixel-wise classification:

$$\mathcal{L}_{CE} = -\sum_{i} y_i \log(\hat{y}_i) \tag{11}$$

where y_i and $\hat{y}_i$ represent ground truth and predicted probabilities, respectively.

3.3.2 Dice Loss

Dice Loss directly optimizes region overlap between predicted and ground-truth masks:

$$\mathcal{L}_{Dice} = 1 - \frac{2\sum_{i=1}^{N}(p_i g_i) + \alpha}{\sum_{i=1}^{N} p_i + \sum_{i=1}^{N} g_i + \alpha} \tag{12}$$

where p_i denotes the predicted foreground probability for pixel i, g_i represents the corresponding ground-truth binary value, N indicates the total number of pixels, and α is a smoothing factor introduced to prevent division by zero [52]. Dice Loss is specifically effective under the conditions of class imbalance, which is commonly observed in weed segmentation datasets.

3.3.3 Boundary Loss

To increase the contour sharpness and localization of the boundaries, we introduce a Boundary Loss based on Laplacian edge detection.

$$E_p = \text{Conv}_{\text{Lap}}(P), \qquad E_g = \text{Conv}_{\text{Lap}}(G) \tag{13}$$

$$\mathcal{L}_{Boundary} = \frac{1}{N}\sum_{i=1}^{N}\left|E_p^i - E_g^i\right| \tag{14}$$

Table 2: Training Details of the Proposed Framework

Batch Size	Epochs	Optimizer	Initial LR	Min LR	Weight Decay	Warmup Epochs	Aux. Loss Weight (λ)	Dropout	Grad. Clip	Aug. Prob.	Input Resolution
8	20	AdamW	2×10^{-4}	1×10^{-6}	1×10^{-3}	3	0.7	0.1	1.0	0.5	512×512

where P and G denote the predicted segmentation map and ground truth mask, E_p and E_g represent their corresponding edge maps, and N is the total number of pixels [53].

3.3.4 Overall Loss

The primary segmentation loss $\mathcal{L}_{\text{main}}$ is defined in the equation 15

$$\mathcal{L}_{main} = 0.4\mathcal{L}_{CE} + 0.4\mathcal{L}_{Dice} + 0.2\mathcal{L}_{Boundary} \tag{15}$$

The $\mathcal{L}_{main}$ enables the model to improve classification accuracy, region overlap, and boundary precision simultaneously.

4 Experiments

To make the comparisons fair, All models were implemented in PyTorch and evaluated on the WeedyRice-RGBMS-DB dataset using the same training, validation, and testing partitions, input resolution, and evaluation metrics. Common preprocessing and augmentation procedures were applied to ensure consistent experimental conditions. The training configuration of the proposed framework is summarized in Table 2. The training process for all models used both RGB and multispectral images, it was performed on a 13th Gen Intel(R) Core i9-13900K (3.00 GHz) processor, an NVIDIA RTX A5000 GPU with 24 GB memory, 128 GB RAM, 1 TB SSD, and 2 TB HDD. The Software environment was Windows 11 Pro, Python 3.10.20, and CUDA 12.4 for accelerated training. The detailed training hyperparameters and optimization settings are presented in Table 2. We conducted a series of experiments on the WeedyRice-RGBMS-DB based on a uniform training and testing scheme. Our algorithm is compared against other state-of-the-art models, such as CNN models (U-Net), transformer models (Vision Transformer and SegFormer), as well as the hybrid models (TransUNet, Swin models, and DeepLabV3+). To thoroughly investigate the proposed model, all methods are evaluated under the same conditions and fairly compared with the standard segmentation metrics, including Accuracy, Precision, Recall, IoU, and Dice coefficient. The results are given as the mean ± standard deviation over several runs to evaluate robustness. Moreover, ablation studies and qualitative comparisons were carried out to analyse the contribution of each component of our proposed model.

4.1 Dataset Description

The experiments are conducted on the Weedy Rice RGB-Multispectral (WeedyRice-RGBMS-DB) dataset [54], a UAV-based multimodal dataset designed for weedy rice segmentation in agricultural fields. The dataset was collected using a DJI M3M UAV over rice fields in An Giang Province, Vietnam, and is geo-referenced in the WGS

Table 3: Dataset Statistics and Split Details

Category	Count
RGB Images	734
Segmentation Masks	734
Overlay Images	734
Multispectral Images (Total)	2936
Multispectral Breakdown (per RGB image)	
Green (G)	734
Near-Infrared (NIR)	734
Red (R)	734
Red Edge (RE)	734
Dataset Split	
Training Set	438
Validation Set	148
Testing Set	148

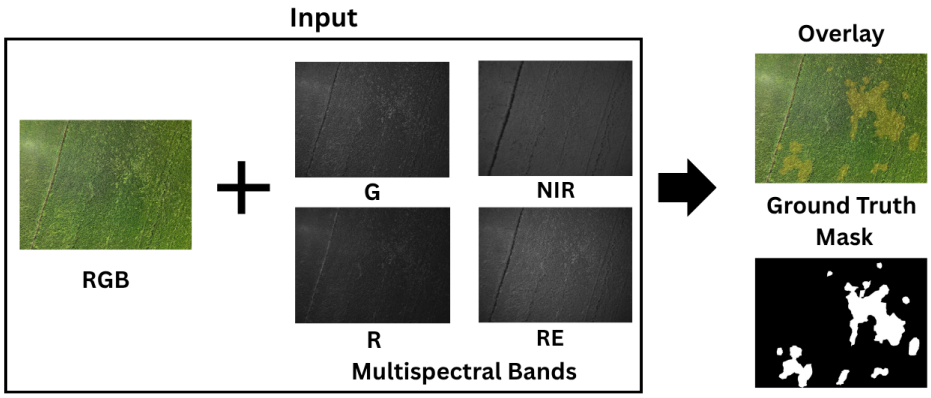


Fig. 3: Overview of the WeedyRice-RGBMS-DB dataset structure. Each sample consists of an RGB image and corresponding multispectral bands, including Green (G), Near-Infrared (NIR), Red (R), and Red Edge (RE) channels. The dataset additionally provides overlay visualizations and pixel-wise ground truth masks for weed segmentation.

84/UTM Zone 48N coordinate system. The imagery was acquired across three cropping seasons. We use the train:validation:test partition (438:148:148) supplied by the dataset authors without modification, so that our results are comparable under a common, publicly defined protocol. The public documentation does not specify the field, flight, or acquisition date of each sample, so we cannot exclude spatial adjacency or temporal continuity between partitions. According to the original dataset publication, the UAV flights were conducted with 70% front and side image overlap to ensure sufficient spatial coverage. This percentage refers to overlap during image acquisition and does not indicate the degree of overlap between training and testing samples. Since the dataset documentation does not establish spatially independent partitions, potential

correlation between neighboring images across the dataset splits cannot be ruled out. It includes RGB images, multispectral images, binary segmentation masks, overlay visualizations, and associated metadata. A sample dataset structure is shown in Fig 3. The dataset consists of a total of 734 samples, divided into training, validation, and test splits. Each sample contains an RGB image (3 channels) along with multispectral bands, namely Green (G), Red (R), Red Edge (RE), and Near-Infrared (NIR), and a corresponding pixel-wise binary ground-truth mask. All images are resized to 512×512 resolution for uniform processing. For multispectral experiments, the model adopts a dual-stream architecture where the RGB encoder receives a 3-channel RGB image and the multispectral encoder receives a 6-channel input composed of 4 spectral bands and 2 vegetation indices.

In terms of dataset composition, the RGB subset contains 734 images of resolution 1280×960, while the multispectral subset includes 2936 spatially aligned band images (four bands per RGB image). The ground-truth masks (734 images) provide pixel-level annotations for weedy rice detection, where pixel value 255 denotes weedy rice and 0 represents the background. Overlay images are provided for qualitative evaluation. Metadata files include acquisition details such as GPS coordinates, altitude, timestamp, sensor type, and filename mappings to ensure data consistency.

4.2 Evaluation Metrics

To quantify crop-weed segmentation result, performance is reported across five key metrics: Accuracy, Precision, Recall, IoU, and Dice score.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{16}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{17}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{18}$$

$$\text{IoU} = \frac{TP}{TP + FP + FN} \tag{19}$$

$$\text{Dice} = \frac{2TP}{2TP + FP + FN} \tag{20}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

5 Results and Discussion

To evaluate the effectiveness of the proposed framework, extensive experiments are conducted against a diverse set of state-of-the-art (SOTA) models, including CNN-based, transformer-based, and hybrid CNN–Transformer architectures. The compared methods include U-Net [12], PSPNet [33], U-Net++ [32], DenseUNet [55], DeepLabV3 [15], Vision Transformer (ViT) [22], SegFormer [23], Swin Transformer-based models [24], CRF-UNet [42], and hybrid CNN-Transformer, frameworks [56].

Table 4: Performance comparison of different models for crop-weed segmentation. Results are reported as mean ± standard deviation (in %).

Method	Accuracy (%)	Precision (%)	Recall (%)	IoU (%)	Dice (%)
U-Net [12]	89.68 ± 0.40	82.18 ± 2.59	80.44 ± 2.36	68.75 ± 0.46	80.97 ± 0.33
Transformer (ViT) [22]	82.48 ± 0.57	66.99 ± 1.94	76.78 ± 2.45	56.08 ± 0.20	70.87 ± 0.19
HRNet [34]	81.67 ± 0.62	63.48 ± 1.88	73.21 ± 2.76	52.72 ± 0.41	67.36 ± 0.43
PSPNet [33]	89.87 ± 0.33	79.82 ± 1.35	83.27 ± 1.61	69.34 ± 0.59	81.36 ± 0.40
DeepLabV3 [15]	88.89 ± 0.24	79.81 ± 2.11	80.23 ± 2.43	66.95 ± 0.54	79.62 ± 0.41
CRF-U-Net [42]	88.91 ± 0.89	80.11 ± 2.01	79.76 ± 3.86	66.82 ± 2.55	79.54 ± 1.93
SegFormer [23]	86.04 ± 0.63	78.65 ± 2.23	70.58 ± 3.54	59.04 ± 1.87	73.13 ± 1.56
CNN + Transformer [56]	88.36 ± 0.52	74.39 ± 1.81	80.59 ± 1.53	63.55 ± 0.65	76.30 ± 0.55
Swin Transformer [24]	90.58 ± 0.39	85.99 ± 2.19	86.13 ± 2.83	75.43 ± 1.03	85.99 ± 0.67
U-Net++ [32]	90.41 ± 0.13	80.28 ± 2.16	78.26 ± 1.81	66.39 ± 0.27	78.56 ± 0.22
DenseUNet [55]	90.75 ± 0.26	83.77 ± 2.88	81.38 ± 2.78	70.67 ± 0.31	82.27 ± 0.23
M^2FA-Net(Proposed)	**94.06 ± 0.07**	**90.72 ± 0.69**	**87.82 ± 0.84**	**80.79 ± 0.24**	**89.21 ± 0.15**

Table 4 presents the quantitative results, where all metrics are reported as mean ± standard deviation (in %) over five independent runs, each with a different seed value. The results show a gradual performance improvement from conventional CNN-based methods to transformer-based and hybrid architectures, highlighting the importance of adaptively combining local spatial information with global contextual representations.

CNN-based architectures such as U-Net provide a strong baseline, achieving 89.68% accuracy and a 68.75% IoU. These models capture local spatial patterns important for distinguishing crop and weed textures. However, their receptive field is limited, which restricts their ability to model long-range contextual dependencies in complex agricultural scenes. Advanced CNN architectures attempt to address this limitation through improved feature aggregation strategies. PSPNet learns the context information of multiple scales by pyramid pooling, U-Net++ improves the feature propagation by nested skip connections, and DenseUNet applies dense connectivity to reuse features more effectively. Thus, DenseUNet achieves better performance than other conventional CNN-based methods with an IoU score of 70.67%, while U-Net++ also provides competitive performance through improved feature propagation.

Transformer-based approaches such as ViT and SegFormer use self-attention mechanisms to better capture global dependencies. However, the local inductive bias of pure transformer architectures is weaker, which limits their ability to preserve the fine spatial details needed for accurate boundary localization. As a result, ViT achieves only 56.08% IoU, while SegFormer improves the performance to 59.04% IoU and 73.13% Dice through hierarchical feature representations. Nevertheless, these models still underperform compared to stronger CNN-based approaches such as DenseUNet. Hybrid CNN-Transformer architectures combine the strengths of both paradigms by jointly learning local texture information and global contextual relationships. This leads to improved segmentation performance, with CNN+Transformer and Swin Transformer achieving IoU scores of 63.55% and 75.43%, respectively. However, their performance remains limited due to insufficient exploitation of multimodal spectral information and the absence of effective feature fusion mechanisms.

DeepLab-based architectures further improve segmentation performance through multi-scale contextual aggregation using dilated convolutions. DeepLabV3 achieves

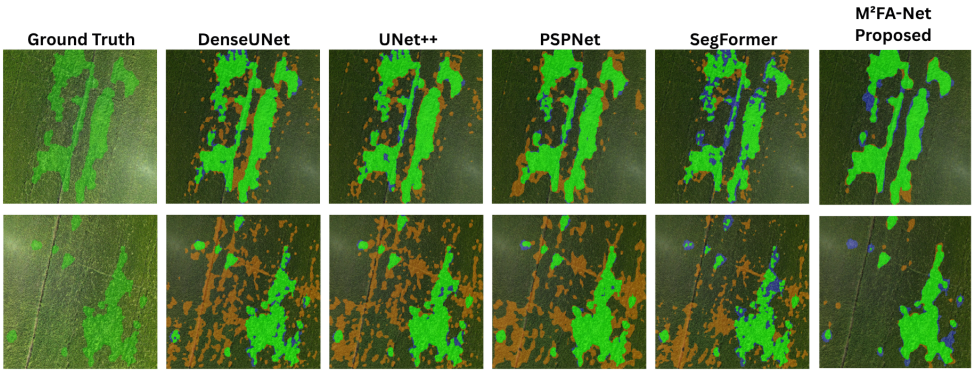


Fig. 4: Qualitative comparison of crop-weed segmentation results produced by different state-of-the-art models on representative field samples. The proposed M^2FA-Net generates cleaner segmentation masks with improved structural consistency, sharper boundary localization, and significantly fewer segmentation artifacts compared to DenseUNet, U-Net++, PSPNet, and SegFormer, particularly in dense vegetation and sparse weed regions.

Table 5: Statistical comparison of M^2FA-Net against selected baselines using two-sided Welch's t-tests on five independent runs. Improvements and confidence intervals are in percentage points. p_{Holm} is adjusted across 10 comparisons (five baselines × two metrics). The 95% confidence intervals are unadjusted individual intervals.

Baseline	IoU			Dice		
	Gain (pp)	**95% CI (pp)**	p_{Holm}	**Gain (pp)**	**95% CI (pp)**	p_{Holm}
U-Net	12.04	[11.47, 12.61]	2.56×10^{-8}	8.24	[7.84, 8.64]	8.11×10^{-8}
SegFormer	21.75	[19.44, 24.06]	4.04×10^{-5}	16.08	[14.15, 18.01]	5.51×10^{-5}
DeepLabV3	13.84	[13.18, 14.50]	8.11×10^{-8}	9.59	[9.09, 10.09]	2.88×10^{-7}
U-Net++	14.40	[14.03, 14.77]	3.88×10^{-12}	10.65	[10.37, 10.93]	4.33×10^{-11}
Swin Transformer	5.36	[4.10, 6.62]	3.87×10^{-4}	3.22	[2.40, 4.04]	3.87×10^{-4}

66.95% IoU and 79.62% Dice, outperforming several CNN and transformer-based approaches. This demonstrates the effectiveness of multi-scale feature extraction and contextual learning for agricultural segmentation tasks. However, convolution-only approaches still face challenges in modeling complex multimodal feature interactions. In contrast, the proposed M^2FA-Net achieves the best performance across all evaluation metrics, obtaining 94.06% accuracy, 90.72% precision, 87.82% recall, 80.79% IoU, and 89.21% Dice score. Compared with Swin Transformer, the strongest competing baseline, the proposed M^2FA-Net improves IoU by 5.36 percentage points and Dice score by 3.22 percentage points. The proposed framework also outperforms DeepLabV3 by 13.84% in IoU and 9.59% in Dice score. In addition, M^2FA-Net has the highest recall among all methods, indicating the reduction of false negatives and improvement of the weed detection capability, which is important for precision agriculture applications.

To examine the reliability of the observed performance gains, we compared M^2FA-Net with U-Net, SegFormer, DeepLabV3, U-Net++, and Swin Transformer using two-sided Welch's t-tests on the mean and sample standard deviation from five independent runs per model. Holm correction was applied across all ten IoU and Dice comparisons. As shown in Table 5, the proposed method achieved statistically significant improvements over all five baselines for both metrics (all Holm-adjusted $p < 0.001$). Against the strongest IoU baseline, Swin Transformer, the improvement was 5.36 percentage points in IoU (95% CI: [4.10, 6.62]) and 3.22 points in Dice (95% CI: [2.40, 4.04]). These findings provide statistical evidence that the observed performance improvements are consistent across repeated training runs on the evaluated test split. However, they do not establish generalization to spatially independent fields or acquisition conditions.

As shown in the first sample in Fig. 4, most baseline methods fail to distinguish weeds from cultivated rice due to the strong visual similarity in terms of texture and color. PSPNet shows aggressive over-segmentation, and SegFormer produces sparse false activations in dense vegetation regions. The proposed M^2FA-Net, on the other hand, produces smoother and more coherent segmentation masks, with sharper boundaries and substantially fewer misclassification errors. The second sample shows a more difficult situation with scarce and dispersed weed regions. Existing approaches are not able to model thin weed structures accurately and tend to classify surrounding background areas as weeds, leading to noisy predictions and more false negatives. In contrast, the proposed M^2FA-Net can accurately detect both large and small weed regions with good boundary preservation and low background noise.

To further examine the proposed framework, Fig. 5 presents additional qualitative examples under challenging field conditions. The model exhibits considerable difficulty in detecting sparse and small isolated weed regions, achieving IoU scores of 23.57% and 48.57%, respectively. These cases reveal missed weed regions and occasional false detections. Dense vegetation also introduces segmentation errors, particularly around irregular boundaries. However, the model maintains relatively accurate predictions in several examples involving uneven illumination and complex canopy structures. These observations highlight both the segmentation capabilities and the limitations of M^2FA-Net under different field conditions.

5.1 Ablation Study of M^2FA-Net

To evaluate the effectiveness of each component in the proposed M^2FA-Net framework, an ablation study is conducted by systematically removing individual modules, including multi-level fusion, CBAM attention, deep supervision, warmup scheduling, and data augmentation, and is given in Table 6. The Baseline (Early Fusion) combines RGB and multispectral inputs before feature extraction using a shared encoder, whereas the No Multi-Level Fusion configuration retains separate modality-specific encoders and combines their features only at the final encoder stage. In contrast, the proposed M^2FA-Net performs hierarchical feature fusion at all four encoder stages. For improved interpretability, the corresponding IoU and Dice trends are additionally visualized in Fig. 6, while Fig. 7 summarizes the relative contribution of each architectural component. As shown in Fig. 6, the complete M^2FA-Net achieves the

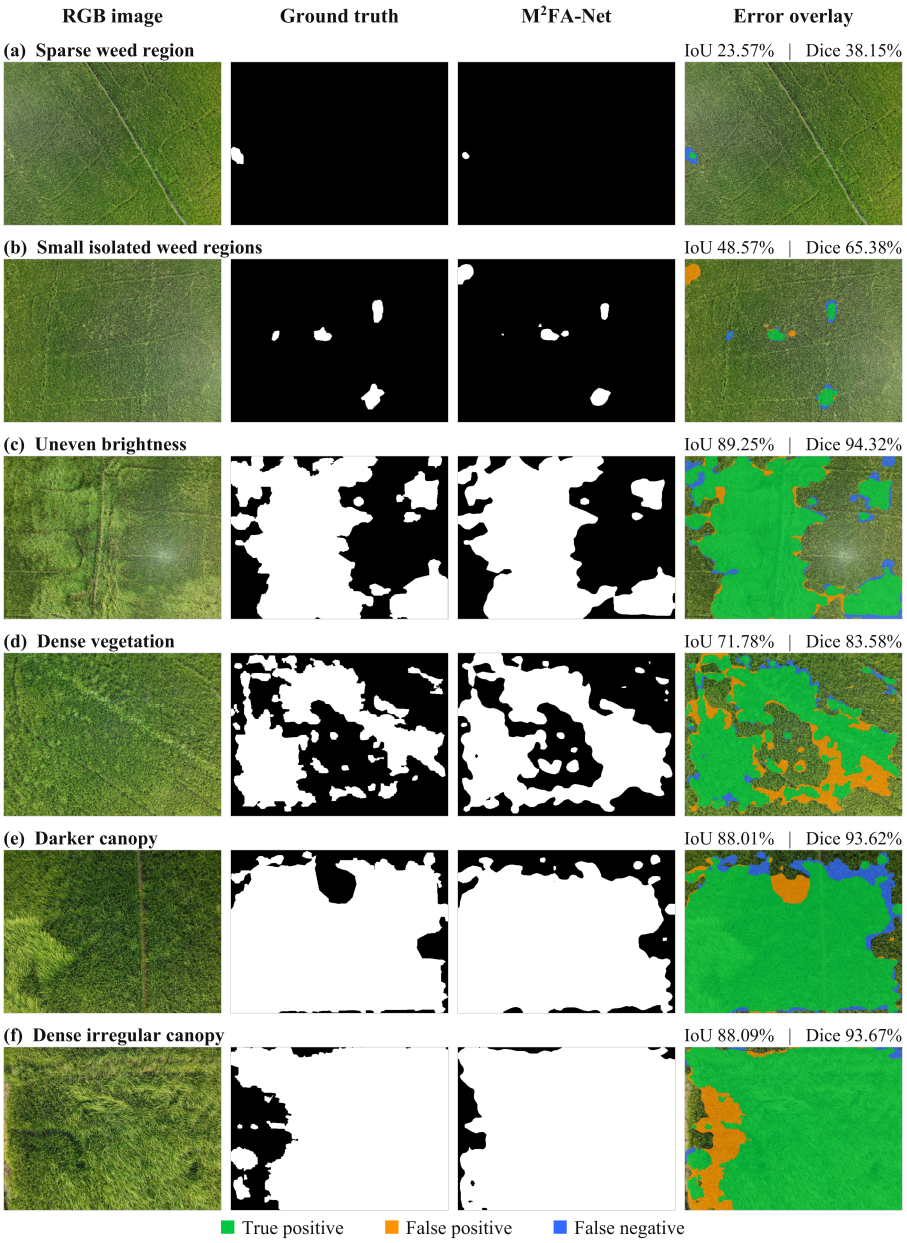


Fig. 5: Qualitative evaluation of M^2FA-Net under challenging field conditions. The examples include sparse weed regions, small isolated weeds, uneven illumination, and dense vegetation. The error overlays distinguish true positives (green), false positives (orange), and false negatives (blue). Per-image IoU and Dice scores are also reported.

Table 6: Ablation study of the architectural and training components of M^2FA-Net. Results are reported as mean ± standard deviation (in %).

Method	Accuracy (%)	Precision (%)	Recall (%)	IoU (%)	Dice (%)
Baseline (Early Fusion)	92.75 ± 0.09	90.28 ± 0.42	82.98 ± 0.43	76.48 ± 0.25	86.41 ± 0.15
No Multi-Level Fusion	92.86 ± 0.09	90.81 ± 0.56	82.80 ± 0.92	76.68 ± 0.42	86.55 ± 0.28
No CBAM Attention	93.10 ± 0.11	90.61 ± 0.80	84.10 ± 0.98	77.62 ± 0.38	87.18 ± 0.25
No Deep Supervision	93.06 ± 0.14	90.95 ± 0.67	83.36 ± 0.99	77.24 ± 0.51	86.92 ± 0.34
No Warmup Scheduler	92.78 ± 0.40	90.92 ± 1.80	83.10 ± 3.08	76.90 ± 1.38	86.72 ± 0.91
No Augmentation	93.26 ± 0.08	90.60 ± 0.28	84.78 ± 0.43	78.18 ± 0.24	87.55 ± 0.15
M^2FA-Net (proposed)	**94.06 ± 0.07**	**90.72 ± 0.69**	**87.82 ± 0.84**	**80.79 ± 0.24**	**89.21 ± 0.15**

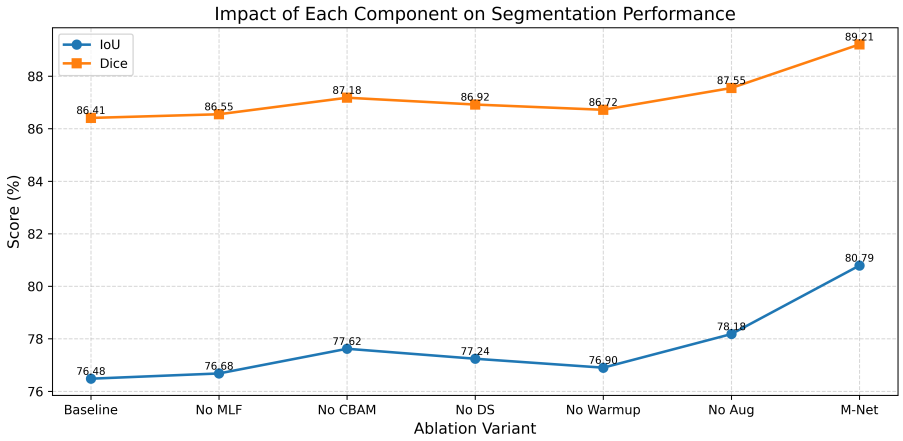


Fig. 6: Comparison of IoU and Dice scores across different ablation configurations of M^2FA-Net.

highest IoU and Dice scores, while removing individual components reduces segmentation performance. The largest reduction occurs when multi-level fusion is removed, highlighting the importance of hierarchical spatial-spectral feature integration. Fig. 7 further illustrates the contributions of the remaining components, particularly deep supervision and CBAM attention.

5.2 Input-Modality Ablation

To quantify the contribution of different input modalities, four configurations are evaluated: (i) RGB only; (ii) RGB with four multispectral bands (MS4: G, R, RE, NIR); (iii) RGB with MS4 and NDVI; and (iv) RGB with MS4, NDVI, and NDRE (the full-input configuration). The RGB-only configuration employs a single RGB encoder, whereas the remaining configurations retain the dual-stream architecture with multi-level feature fusion and the corresponding multispectral input channels. Within this ablation study, the loss function, training schedule, data augmentation, and auxiliary supervision are kept consistent across the configurations. Results are reported as mean ± standard deviation over five runs with different random seeds. Fig. 8 presents the results.

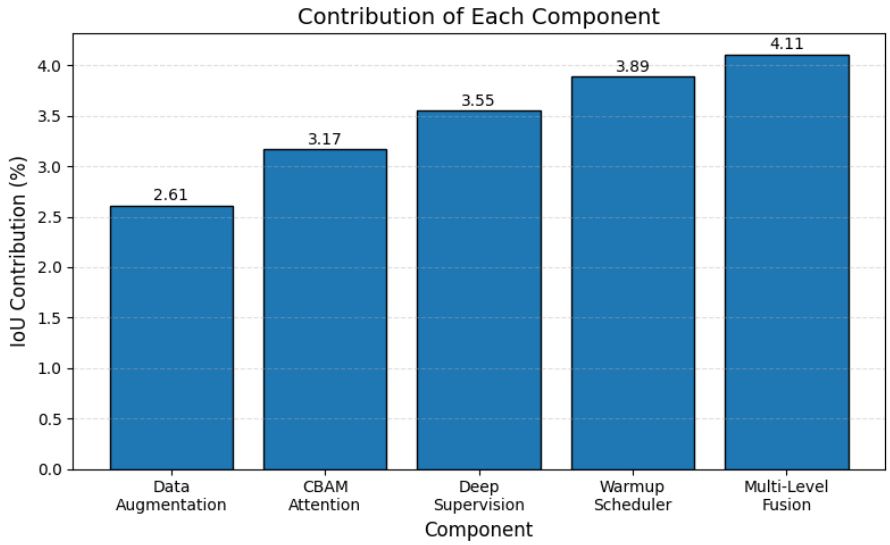


Fig. 7: Relative contribution of individual components to the IoU performance of M^2FA-Net.

Adding the four multispectral bands raises IoU from 83.11% to 83.75% (+0.64 points) and Dice from 90.77% to 91.15% (+0.38 points). The gain is small in absolute terms, but it comes with a marked reduction in run-to-run variability: the IoU standard deviation drops from 1.24 to 0.10. Adding NDVI changes IoU by −0.07 points, and adding NDRE on top of NDVI changes it by +0.13 points; both changes are smaller than the standard deviation of the corresponding runs. The vegetation indices therefore give no measurable improvement beyond the raw bands in our setting, which is plausible because both indices are deterministic functions of bands that the network already receives. Since the RGB-only runs vary more than the gap to the multispectral settings, we regard the improvement from the bands as consistent in direction but modest, and we do not claim a statistically significant difference.

The input-modality ablation was conducted as a separate experimental analysis using the same dataset partitions, loss function, and training duration as the main experiments. However, minor implementation differences existed in the initialization of the multispectral encoder and checkpoint selection. Specifically, the input-modality experiments evaluated the best validation-selected checkpoints, whereas the main experiments evaluated the final-epoch models. Consequently, the absolute performance values of the full-input configuration differ between the two experiments. The contribution of each input modality is therefore assessed through comparisons within the consistent input-modality ablation protocol.

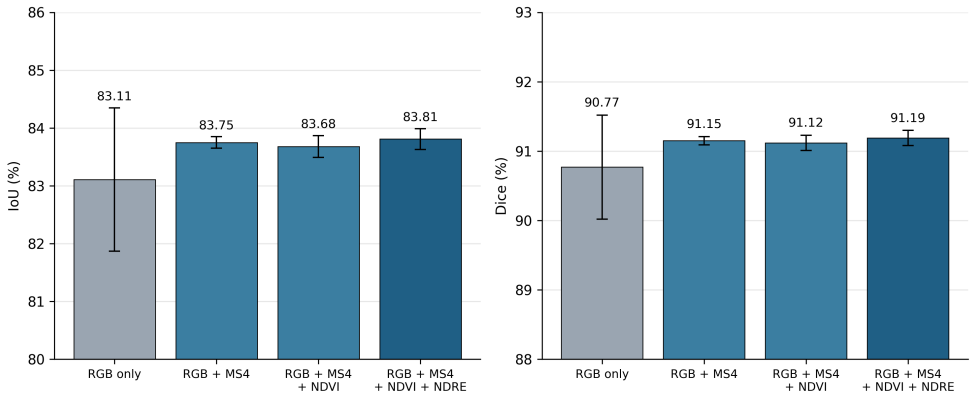


Fig. 8: Input-modality ablation: IoU (left) and Dice (right) for the four input settings. Bars show the mean and error bars the standard deviation over repeated runs; the vertical axes are truncated to make small differences visible.

5.3 Computational Complexity Analysis

To further evaluate the efficiency of the proposed framework, a computational complexity analysis was conducted in terms of the number of parameters, floating point operations (FLOPs), and inference speed measured in frames per second (FPS). All complexity metrics were computed using an input resolution of 512×512, while FPS was measured using batch size 1 under inference mode using the same hardware configuration described in Section IV.

Table 7: Computational complexity comparison of different segmentation models in terms of FLOPs and inference speed

Model	FLOPs (G)	Inference Speed(FPS)
DenseUNet	68.60	47.00
U-Net++	148.29	40.81
PSPNet	19.53	280.11
SegFormer	6.77	194.58
M^2FA-Net (Proposed)	267.68	40.22

Table 7 presents the computational complexity comparison of different segmentation models in terms of FLOPs and inference speed. Figure 9 further compares the parameter count of representative segmentation architectures. Although the proposed framework is computationally more demanding than lightweight architectures such as PSPNet and SegFormer, it still maintains competitive inference speed while delivering significantly superior segmentation accuracy. Taking into account substantial improvements in IoU and Dice score, the additional computational cost is justified

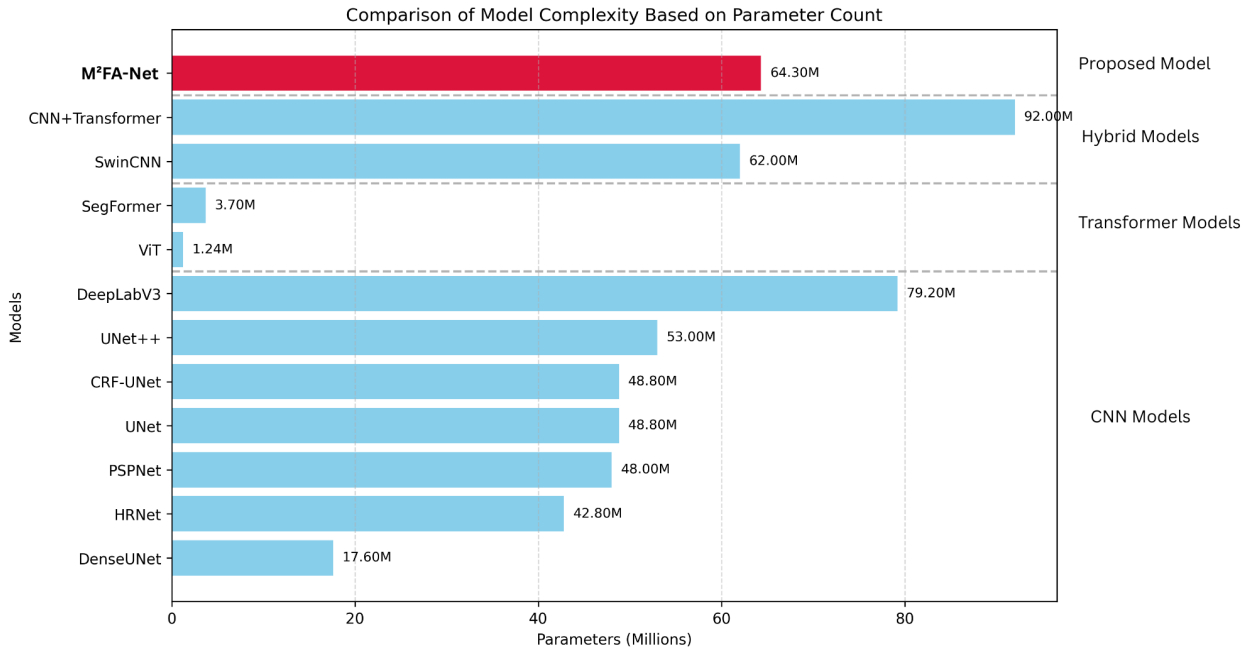


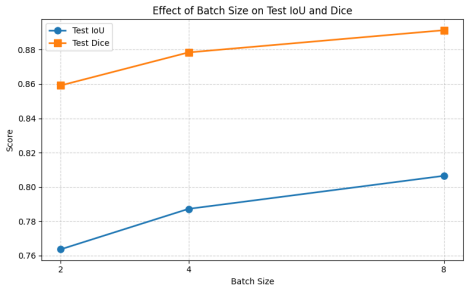
Fig. 9: Comparison of model complexity in terms of parameter count across representative CNN-based, transformer-based, hybrid, and proposed architectures. Although M^2FA-Net contains more parameters than several baseline models, it remains substantially lighter than some transformer-hybrid frameworks while achieving superior segmentation performance.

for precision agriculture applications where segmentation quality is of primary importance. Overall, the proposed M^2FA-Net achieves improved segmentation performance at the cost of increased computational complexity. Therefore, further optimization and hardware-specific evaluation are necessary before considering deployment on resource-constrained UAV platforms.

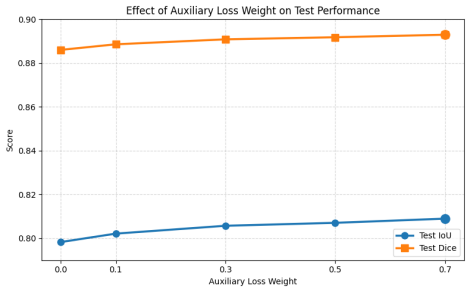
The reported inference speed of 40.22 FPS was obtained on a desktop NVIDIA RTX A5000 GPU and should not be interpreted as evidence of real-time inference capability on UAV or edge devices. Future work will investigate device-specific inference latency, GPU memory consumption, and deployment feasibility on platforms such as NVIDIA Jetson.

5.4 Hyperparameter Sensitivity Analysis

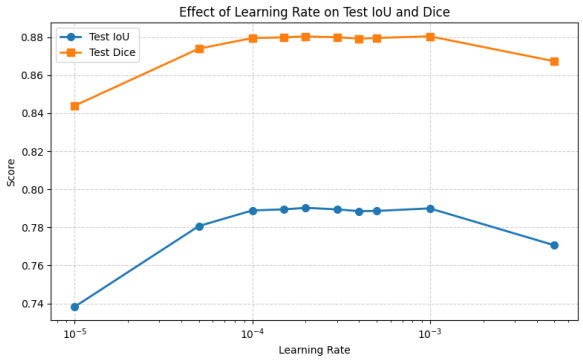
To further investigate the robustness of the proposed M^2FA-Net framework, a series of hyperparameter sensitivity experiments is conducted. Key training parameters, including learning rate, batch size, auxiliary loss weight, warmup epochs, weight decay, and loss function weights, were systematically varied while keeping all other settings unchanged. Fig. 10a reveals the effects of batch sizes on segmentation results. The



(a) Batch size sensitivity.



(b) Auxiliary loss weight sensitivity.

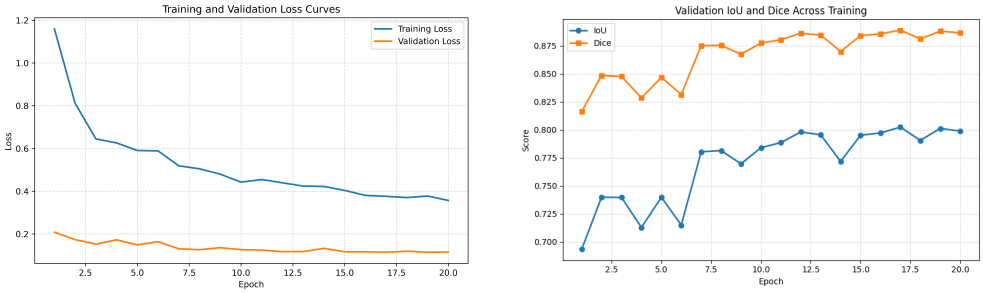


(c) Learning rate sensitivity.

Fig. 10: Hyperparameter sensitivity analysis of the proposed M^2FA-Net.

highest performance among the evaluated settings is achieved with a batch size of 8, which was adopted for all subsequent experiments. Figure 10b shows that increasing the auxiliary loss weight consistently improves segmentation performance. The best performance is achieved at $\lambda = 0.7$, which was adopted in the final M^2FA-Net framework. As illustrated in Fig. 10c, the results improve significantly when increasing the learning rate from 1×10^{-5}, reaching its highest performance at 2×10^{-4}, which was selected for the final configuration of the model.

According to the obtained data, weight decay, warmup epochs, and loss weights have only a slight effect on segmentation efficiency in comparison with the learning rate, batch size, and auxiliary loss weight. For the weight decay parameter within its range from 10^{-5} to 1, the difference in val IoU did not exceed 0.5%. Likewise, variations in the number of warmup epochs between 0 and 5 had only a negligible effect on the framework's performance; the warmup stage of 3 epochs was finally used. Besides, several different loss function weightings for CE, Dice, and Boundary losses led to almost similar results and proved the framework's insensitivity to the specified parameter. Among them, the combination with weights of 0.4 for CE, 0.4 for Dice, and 0.2 for Boundary was recognized as the best one.



(a) Training and validation losses.

(b) Validation IoU and Dice scores.

Fig. 11: Training dynamics of M^2FA-Net over 20 epochs.

To analyze the optimization behavior of the proposed framework, the evolution of training and validation losses is monitored throughout the training process. As shown in Fig. 11a, both losses decrease steadily across epochs, indicating stable convergence and effective optimization. Furthermore, the small gap between the training and validation curves suggests a relatively small difference between training and validation losses under the evaluated data partition. However, these curves alone do not establish generalization to spatially independent field observations. In addition to loss convergence, segmentation performance is tracked using validation IoU and Dice metrics. As illustrated in Fig. 11b, both metrics improve consistently throughout training before stabilizing during the later epochs. The steady increase in validation performance confirms the effectiveness of the proposed training strategy and optimization settings.

6 Conclusion and Future scope

In this paper, we proposed M^2FA-Net, a multimodal fusion framework for crop–weed segmentation that integrates dual-stream DeepLabV3 backbones, multi-level feature fusion, CBAM attention, deep supervision, and a composite loss function. The proposed framework effectively combines spatial, spectral, and vegetation-index information, achieving 80.79% IoU and 89.21% Dice score on the WeedyRice-RGBMS-DB dataset, outperforming the evaluated baseline models. These results demonstrate the effectiveness of M^2FA-Net on the evaluated benchmark. Input-modality ablation further indicates a modest but consistent improvement from multispectral bands, while NDVI and NDRE provide no clear additional benefit.

This work has several limitations. Experiments were conducted on a single dataset of 734 samples collected using one sensor in An Giang, Vietnam, with the image-level split provided by the dataset authors. Although the images were acquired across multiple cropping seasons, the documentation does not confirm whether the partitions are independent by field, UAV flight, or acquisition date. Consequently, spatially or temporally adjacent images may occur across training and test partitions, potentially leading to optimistic performance estimates. The reported results therefore primarily reflect generalization within this dataset. Furthermore, the modest contribution of

multispectral bands and vegetation indices may vary across crops, growth stages, and sensors. The model's computational requirements and lack of edge-device evaluation also limit conclusions about real-time onboard UAV deployment.

Future work will focus on improving generalization, computational efficiency, and practical deployment. Few-shot learning, domain adaptation, domain generalization, and self-supervised multimodal pretraining may improve performance across different agricultural conditions while reducing annotation requirements. Lightweight variants will be explored for deployment on UAVs and edge devices, along with extensions to multi-class segmentation and spatiotemporal analysis. Independent cross-field, cross-season, and external-dataset evaluations will also be conducted to assess generalization more rigorously.

Author Contributions

Manan Dudeja: Methodology, Software, Data curation, Validation, Writing–original draft. Rakesh Kumar Sanodiya: Origination and Conceptualization, Data curation, Validation, Supervision, Project administration, Investigation, Writing–review & editing. Lekshmi R: Formal analysis, Visualization, Investigation. Matloob Khushi: Conceptualization, Validation, Supervision, Project administration, Writing–review & editing.

Acknowledgment

We thank the anonymous reviewers for their careful reading and constructive feedback, which helped improve this manuscript.

Funding

This work was supported by the Science and Engineering Research Board (SERB) under the Core Research Grant Project (File No. CRG/2023/001239).

Data Availability

The WeedyRice-RGBMS-DB dataset used in this work is publicly available at: https://data.mendeley.com/datasets/vt4s83pxx6/1

Competing Interests

The authors declare that they have no competing interests.

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